JIE ZHANG

tel: +8619910031394|jiez@pku.edu.cn

Dr. Jie Zhang is currently an assistant professor of the school of Computer Science at Peking University, China. He is engaged in the research and design of storage systems, non-volatile memory and specialized processors. His research addresses the requirements for high-performance storage systems in the era of big data and artificial intelligence from the perspective of computer architecture. He is dedicated to breaking through the bottlenecks of data migration and the limitations of memory walls in the Von Neumann architecture. The projects he leads and have participated in have been funded by the Natural Science Foundation of China, Natural Science Foundation of Beijing, National Key Research and Development Program of China, Alibaba, Ant Research and Longsys Inc. with a cumulative total of over 4 million dollars. He has published over 60 papers in top-tier conferences and journals, including ISCA, OSDI, SOSP, HPCA, MICRO, ASPLOS, FAST, ATC, and Eurosys. **Among post-90s scholars in computer architecture, he ranks first globally in the number of papers published in top conferences (CSRankings, 2016–2026).** He was awarded **ACM SIGCSE Rising Star**, **Honor Scholar of Intel China Academic Talent Program** and **OlympusMons Award**. He was inducted to HPCA Hall of Fame in 2025.

WORK EXPERIENCE

Peking University, Beijing, China — **Assistant Professor, PhD advisor**
School of Computer science — July 2021 –Current

KAIST, Daejeon, Korea — **Postdoctoral Researcher**
Electrical Engineering (Computing Division) — March 2020 –May 2021

EDUCATION

Yonsei University, Incheon, Korea — **Advisor:** Dr. Myoungsoo Jung
PhD in Engineering — August 2015 – Feb 2020

University of Texas at Dallas, Richardson, Texas — **Advisor:** Dr. Myoungsoo Jung
PhD in Computer Engineering (transfer to Korea) — August 2014 – August 2015

University of Texas at Dallas, Richardson, Texas — **Advisor:** Dr. Myoungsoo Jung
Master of Science in Electrical Engineering — August 2012 – May 2014

Nanjing University of Posts and Telecommunications, Nanjing, China
BS in Communication Engineering (computer communication) — September 2008 – July 2012

PUBLICATIONS

2026

ICS — **Latency-SLO-Aware Memory Offloading for Large Language Model Inference**

Top Conference — *Chenxiang Ma, Hanyu Zhao, Zhisheng Ye, Zehua Yang, Tianhao Fu, Jiaxun Han, **Jie Zhang**, Yingwei Luo, Xiaolin Wang, Zhenlin Wang, Yong Li, Diyu Zhou*

ACM International Conference on Supercomputing

HPDC — **BCCE: Block-Centric GPU Co-Design for Real-Time Range-Top-K Query at Scale**

Top Conference — *Chengying Huan, Ziheng Meng, Zhengyi Yang, Yongchao Liu, **Jie Zhang**, Qing Wang, Jing Wang, Shaonan Ma, Zhibin Wang, Mingxing Zhang, Rong Gu, Baokun Wang, Guihai Chen, Chen Tian*

ACM International Symposium on High-Performance Parallel and Distributed Computing

OSDI — **Espresso: Constructing Cost-Efficient CXL JBOF via Inter-SSD Computing Resource**

JIE ZHANG

tel: +8619910031394|jiez@pku.edu.cn

Top Conference **Sharing**

*Shushu Yi, Yuda An, Li Peng, Xiurui Pan, Qiao Li, Jieming Yin, Guangyan Zhang, Wenfei Wu, Chenxi Wang, Diyu Zhou, Zhenlin Wang, Xiaolin Wang, Yingwei Luo, Ke Zhou, **Jie Zhang****

USENIX Symposium on Operating Systems Design and Implementation

OSDI **Merlin: An Efficient Adaptive Cache Eviction Algorithm via Fine-Grained Characterization**

Top Conference *Liujia Li, Jinhao Guo, Yi Fan, Jianyu Wu, Zhenling Wang, **Jie Zhang**, Yuval Tamir, Xiaolin Wang, Yingwei Luo, Diyu Zhou*

USENIX Symposium on Operating Systems Design and Implementation

OSDI **MultiLane: Eliminating Centralized Bottlenecks in User-space Network Stack**

Top Conference *Kang Hu, Shuqi Dong, Chuandong Li, Ran Yi, Zonghao Zhang, **Jie Zhang**, Yingwei Luo, Xiaolin Wang, Zhenling Wang, Diyu Zhou*

USENIX Symposium on Operating Systems Design and Implementation

ICLR **LouisKV: Efficient KV Cache Retrieval for Long Input-Output Sequences**

Top Conference *Wenbo Wu, Qingyi Si, Xiurui Pan, Ye Wang, **Jie Zhang****

The Fourteenth International Conference on Learning Representations

DAC **WILL: Write Invalidation Skipping for QLC NAND by Leveraging Data Lifespan and Latency Awareness**

Top Conference *Beichen Ning, Yina Lv, Tianyu Ren, Bin Gao, Xiangyu Yao, **Jie Zhang**, Qiao Li and Chun Jason Xue*

Design Automation Conference

DAC **Cachence: Fine-Grained Cache Partitioning in Both Time and Space**

Top Conference *Liujia Li, Yuanlong Li, Yiming Yao, Jianyu Wu, Yi Fan, Jinhao Guo, Liren Zhu, **Jie Zhang**, Xiaolin Wang, Yingwei Luo, Zhenlin Wang, Diyu Zhou*

Design Automation Conference

FAST **Xerxes: Extensive Exploration of Scalable Hardware Systems with CXL-Based Simulation**

Top Conference **Framework**

*Yuda An, Shushu Yi, Bo Mao, Qiao Li, Mingzhe Zhang, Diyu Zhou, Ke Zhou, Nong Xiao, Guangyu Sun, Yingwei Luo, **Jie Zhang****

The USENIX Conference on File and Storage Technologies

HPCA **AutoGNN: End-to-End Hardware-Driven Graph Preprocessing for Enhanced GNN**

Top Conference **Performance**

*Seungkwan Kang, Seungjun Lee, Donghyun Gouk, Miryeong Kwon, Hyunkyu Choi, Junhyeok Jang, Sangwon Lee, Huiwon Choi, **Jie Zhang**, Wonil Choi, Mahmut T Kandemir, Myoungsoo Jung*

IEEE International Symposium on High-Performance Computer Architecture

Eurosys **ColdCode: Cold Data Encoding for Enhanced Reliability and Lifetime in 3D NAND Flash**

Top Conference *Qiao Li, Shangyu Wu, Zheng Wan, Yufei Cui, **Jie Zhang***, Chun Jason Xue*

21th ACM European Conference on Computer Systems

JIE ZHANG

tel: +8619910031394|jiez@pku.edu.cn

2025

TCAD **CXL-DMSim: A Full-System CXL Disaggregated Memory Simulator With Comprehensive Silicon Validation**

Top Journal *Yanjing Wang, Lizhou Wu, Wentao Hong, Yang Ou, Zicong Wang, Sunfeng Gao, **Jie Zhang**, Sheng Ma, Dezun Dong, Xingyun Qi, Mingche Lai, Nong Xiao*

PACT **Exploring Memory Tiering Systems in the CXL Era via FPGA-based Emulation and Device-**

Top Conference **Side Management**

*Yiqi Chen, Xiping Dong, Zhe Zhou, Zhao Wang, **Jie Zhang**, Guangyu Sun,*

International Conference on Parallel Architecture and Compilation Techniques

NEWTON **Ultrafast and Reliable Domain-Wall and Skyrmion Logic in a Chirally Coupled Ferrimagnet**

Top Journal *Yifei Ma, Dihua Wu, Fengbo Yan, Xiaoxiao Fang, Peixin Qin, Leran Wang, Li Liu, Laichuan Shen, Zhiqi Liu, Wenyun Yang, **Jie Zhang**, Yanglong Hou, Yan Zhou, Feng Luo, Jinbo Yang, Kaiming Cai, Shuai Ning, Zhaochu Luo*

Cell Press NEWTON

SOSP **Aeolia: Fast and Secure Userspace Interrupt-Based Storage Stack**

Top Conference *Chuandong Li, Ran Yi, Zonghao Zhang, Jing Liu, Changwoo Min, **Jie Zhang**, Yingwei Luo, Xiaolin Wang, Zhenling Wang, Diyu Zhou*

ACM Symposium on Operating Systems Principles

HotStorage **SPDK+: Low Latency or High Power Efficiency? We Take Both**

Endian Li, Shushu Yi, Li Peng, Qiao Li, Diyu Zhou, Zhenlin Wang, Xiaolin Wang, Bo Mao,

*Yingwei Luo, Ke Zhou, **Jie Zhang****

The 17th ACM Workshop on Hot Topics in Storage and File Systems

ICDE **BL-Tree: The Best of Both Worlds by combining B+-Tree on Top and LSM-Tree on Bottom**

Top Conference *Maoxin Ye, Zuocheng Wang, Shengzhe Wang, Jiahong Chen, Chunfeng Du, **Jie Zhang**, Ke Zhou, Suzhen Wu, Bo Mao*

IEEE International Conference on Data Engineering

ToS **Enhancing the Performance of Next-Generation SSD Arrays: A Holistic Approach**

Top Journal ***Jie Zhang****, *Shushu Yi, Xiurui Pan, Qiao Li, Qiang Li, Chenxi Wang, Bo Mao, Myoungsoo Jung*

ACM Transactions on Storage

TCAD **FlexHMB: A Flexible HMB Design towards Bufferless Mobile Flash**

Top Journal *Yong Peng, Li Peng, Shaocong Sun, Lujia Yin, Qiao Li, **Jie Zhang****

ACM Transactions on Computer-Aided Design of Integrated Circuits and Systems

ISCA **XHarvest: Rethinking High-Performance and Cost-Efficient SSD Architecture with**

Top Conference **CXL-Driven Harvesting**

*Li Peng, Wenbo Wu, Shushu Yi, Xianzhang Chen, Chenxi Wang, Shengwen Liang, Zhe Wang, Nong Xiao, Qiao Li, Mingzhe Zhang, **Jie Zhang****

IEEE/ACM International Symposium on Computer Architecture

ISCA — **ArtMem: Adaptive Migration in Reinforcement Learning-Enabled Tiered Memory**

Top Conference — *Xinyue Yi, Hongchao Du, Yu Wang, **Jie Zhang**, Qiao Li, Chun Jason Xue*

IEEE/ACM International Symposium on Computer Architecture

Eurosys — **Daredevil: Rescue Your Flash Storage from Inflexible Kernel Storage Stack**

Top Conference — *Junzhe Li, Ran Shu, Jiayi Lin, Qingyu Zhang, Ziyue Yang, **Jie Zhang**, Yongqiang Xiong, Chenxiong Qian*

20th ACM European Conference on Computer Systems

NSDI — **Beehive: A Scalable Disaggregated Memory Runtime Exploiting Asynchrony of Multithreaded Programs**

Top Conference — *Quanxi Li, Hong Huang, Ying Liu, Yanwen Xia, **Jie Zhang**, Mosong Zhou, Xiaobing Feng, Huimin Cui, Quan Chen, Yizhou Shan, Chenxi Wang*

22nd USENIX Symposium on Networked Systems Design and Implementation

HPCA — **InstAttention: In-Storage Attention Offloading for Cost-Effective Long-Context LLM Inference**

Top Conference — *Xiurui Pan, Endian Li, Qiao Li, Shengwen Liang, Yizhou Shan, Ke Zhou, Yingwei Luo, Xiaolin Wang, **Jie Zhang***

IEEE International Symposium on High-Performance Computer Architecture

HPCA — **NeuVSA: A Unified and Efficient Accelerator for Neural Vector Search**

Top Conference — *Ziming Yuan, Lei Dai, Wen Li, **Jie Zhang**, Shengwen Liang, Ying Wang, Cheng Liu, Huawei Li, Jiafeng Guo, Peng Wang, Renhai Chen, Gong Zhang*

IEEE International Symposium on High-Performance Computer Architecture

HPCA — **Criticality-Aware Instruction-Centric Bandwidth Partitioning for Data Center Applications**

Top Conference — *Liren Zhu, Liujia Li, Jianyu Wu, Yiming Yao, Zhan Shi, **Jie Zhang**, Zhenlin Wang, Xiaolin Wang, Yingwei Luo, Diyu Zhou*

IEEE International Symposium on High-Performance Computer Architecture

2024

SOSP — **BIZA: Design of Self-Governing Block-Interface ZNS AFA for Endurance and Performance**

Top Conference — *Shushu Yi, Shaocong Sun, Li Peng, Yingbo Sun, Ming-Chang Yang, Zhichao Cao, Qiao Li, Myoungsoo Jung, Ke Zhou, **Jie Zhang***

ACM Symposium on Operating Systems Principles

MICRO — **NeoMem: Hardware/Software Co-Design for CXL-Native Memory Tiering**

Top Conference — *Zhe Zhou, Yiqi Chen, Tao Zhang, Yang Wang, Ran Shu, Shuotao Xu, Peng Cheng, Lei Qu, Yongqiang Xiong, **Jie Zhang**, Guangyu Sun*

The 57th Annual IEEE/ACM International Symposium on Microarchitecture

MICRO — **Cambricon-llm: A chiplet-based hybrid architecture for on-device Inference of 70B LLM**

Top Conference — *Zhongkai Yu, Shengwen Liang, Tianyu Ma, Yunke Cai, Ziyuan Nan, Di Huang, Xinkai Song, Yifan Hao, **Jie Zhang**, Tian Zhi, Yongwei Zhao, Zidong Du, Xing Hu Qi Guo, Tianshi Chen*

The 57th Annual IEEE/ACM International Symposium on Microarchitecture

ToS — **Extremely-Compressed SSDs with I/O Behavior Prediction**

Top Journal — *Xiangyu Yao, Qiao Li, Kaihuan Lin, Xinbiao Gan, **Jie Zhang**, Congming Gao, Zhirong Shen, Quanqing Xu, Chuanhui Yang, Jason Xue*

ACM Transactions on Storage

TC — **Land of Oz: Resolving Orderless Writes in Zoned Namespace SSDs**

Top Journal — *Yingjia Wang, You Zhou, Fei Wu, **Jie Zhang**, Ming-Chang Yang*

IEEE Transactions on Computers

USENIX ATC — **ScalaAFA: Constructing User-Space All-Flash Array Engine with Holistic Designs**

Top Conference — *Shushu Yi, Xiurui Pan, Qiao Li, Qiang Li, Chenxi Wang, Bo Mao, Myoungsoo Jung, **Jie Zhang***

USENIX Annual Technical Conference

USENIX ATC — **ScalaCache: Scalable User-Space Page Cache Management with Software-Hardware Coordination**

Top Conference

*Li Peng, Yuda An, You Zhou, Chenxi Wang, Qiao Li, Chuanning Chen, **Jie Zhang***

USENIX Annual Technical Conference

Arxiv — **From RDMA to RDCA: Toward High-Speed Last Mile of Data Center Networks Using Remote Direct Cache Access**

*Qiang Li, Qiao Xiang, Derui Liu, Yuxin Wang, Haonan Qiu, Xiaoliang Wang, **Jie Zhang**, Ridi Wen, Haohao Song, Gexiao Tian, Chenyang Huang, Lulu Chen, Shaozong Liu, Yaohui Wu, Zhiwu Wu, Zicheng Luo, Yuchao Shao, Chao Han, Zongjie WU, Jianbo Dong, Zheng Cao, Jinbo WU, Jiwu Shu, Jiesheng Wu*

Arxiv

TACO — **Characterizing and Optimizing LDPC Performance on 3D NAND Flash Memories**

Top Journal — *Qiao Li, Yu Chen, Guanyu Wu, Yajuan Du, Min Ye, Xinbiao Gan, **Jie Zhang**, Zhirong Shen, Jiwu Shu, Chun Jason Xue*

ACM Transactions on Architecture and Code Optimization

ISCA — **Flagger: Cooperative Acceleration for Large-Scale Cross-Silo Federated Learning Aggregation**

Top Conference

*Xiurui Pan, Yuda An, Shengwen Liang, Bo Mao, Mingzhe Zhang, Qiao Li, Myoungsoo Jung, **Jie Zhang***

IEEE/ACM International Symposium on Computer Architecture

ASPLOS — **Achieving Near-Zero Read Retry for 3D NAND Flash Memory**

Top Conference — *Min Ye, Qiao Li, Yina Lv, **Jie Zhang**, Tianyu Ren, Daniel Wen, Tei-Wei Kuo, Chun Jason Xue*

ACM International Conference on Architectural Support for Programming Languages and Operating Systems

HPCA — **StreamPIM: Streaming Matrix Computation in Racetrack Memory**

Top Conference — *Yuda An, Yunxiao Tang, Shushu Yi, Li Peng, XIurui Pan, Guangyu Sun, Zhaochu Luo, Qiao Li, **Jie Zhang***

IEEE International Symposium on High-Performance Computer Architecture

HPCA **BeaconGNN: Large-Scale GNN Acceleration with Asynchronous In-Storage Computing**

Top Conference *Yuyue Wang, Xiurui Pan, Yuda An, **Jie Zhang****, *Glenn Reinman**

IEEE International Symposium on High-Performance Computer Architecture

HPCA **LearnedFTL: A Learning-based Page-level FTL for Reducing Double Reads in Flash-based**

Top Conference **SSDs**

*Shengzhe Wang, Zihang Lin, Suzhen Wu, Hong Jiang, **Jie Zhang**, Bo Mao*

IEEE International Symposium on High-Performance Computer Architecture

HPCA **Midas Touch: Invalid-Data Assisted Reliability and Performance Boost for 3D High-**

Top Conference **Density Flash**

*QiaoLi, Hongyang Dang, Zheng Wan, Congming Gao, Min Ye, **Jie Zhang**, Tei-Wei Kuo, Chun Jason Xue*

IEEE International Symposium on High-Performance Computer Architecture

2023

NVMW **Optimizations of Linux Software RAID System for Next-Generation Storage**

*Shushu Yi, Yanning Yang, Yunxiao Tang, Zixuan Zhou, Junzhe Li, Yue Chen, Myoungsoo Jung, **Jie Zhang****,

The 14th Annual Non-volatile Memories Workshop

SAC **BcBench: Exploring Throughput Processor Designs based on Blockchain Benchmarking**

*Xiurui Pan, Yue Chen, Shushu Yi, **Jie Zhang****,

The 38th ACM/SIGAPP Symposium on Applied Computing

CAL **Intelligent SSD Firmware for Zero-Overhead Journaling**

SCI Journal *Hanyeoreum Bae, Donghyun Gouk, Seungjun Lee, Jiseon Kim, Sungjoon Koh, **Jie Zhang****, *and Myoungsoo Jung**,

IEEE Computer Architecture Letters (CAL)

2022

HotStorage **ScalaRAID: Optimizing Linux Software RAID System for Next-Generation Storage**

*Shushu Yi, Yanning Yang, Yunxiao Tang, Zixuan Zhou, Junzhe Li, Yue Chen, Myoungsoo Jung, **Jie Zhang****,

14th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage 22)

THPC **Survey on Storage-Accelerator Data Movement**

*Zixuan Zhou, Shushu Yi, **Jie Zhang****,

CCF Transaction on High Performance Computing

NVMW **Integrating New Photonic-Based Heterogeneous Memory into Throughput Accelerators**

***Jie Zhang**, Myoungsoo Jung,*

The 13th Annual Non-volatile Memories Workshop

NVMW	**HAMS: Hardware Automated Memory-over-Storage for Large-scale Memory Expansion**
	Jie Zhang, *Miryeong Kwon, Donghyun Gouk, Sungjoon Koh, Nam Sung Kim,*
	Mahmut Taylan Kandemir, Myoungsoo Jung
	The 13th Annual Non-volatile Memories Workshop

2021

MICRO	**Ohm-GPU: Integrating New Optical Network and Heterogeneous Memory into GPU**
Top Conference	**Multi-Processors**
	Jie Zhang, *Myoungsoo Jung,*
	The 54th Annual IEEE/ACM International Symposium on Microarchitecture
ISCA	**Revamping Storage Class Memory with Hardware Automated Memory-Over-Storage**
Top Conference	**Solution**
	Jie Zhang, *Miryeong Kwon, Donghyun Gouk, Sungjoon Koh, Nam Sung Kim,*
	Mahmut Kandemir, Myoungsoo Jung,
	The IEEE/ACM International Symposium on Computer Architecture
NVMW	**Architecting Throughput Processors with New Flash**
	Jie Zhang, *Myoungsoo Jung,*
	The 12th Annual Non-volatile Memories Workshop
NVMW	**DRAM-less Accelerator for Energy Efficient Data Processing**
	Jie Zhang, *Gyuyoung Park, David Donofrio, John Shalf, Myoungsoo Jung*
	The 12th Annual Non-volatile Memories Workshop
NVMW	**Manycore-Based Scalable SSD Architecture Towards One and More Million IOPS**
	Jie Zhang, *Miryeong Kwon, Michael Swift, Myoungsoo Jung,*
	The 12th Annual Non-volatile Memories Workshop

2020

ISCA	**ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis**
Top Conference	***Jie Zhang***, *Myoungsoo Jung,*
	The IEEE/ACM International Symposium on Computer Architecture
FAST	**Scalable Parallel Flash Firmware for Many-core Architectures**
Top Conference	***Jie Zhang***, *Miryeong Kwon, Michael Swift, Myoungsoo Jung,*
	The 18th USENIX Conference on File and Storage Technologies
HPCA	**DRAM-less: Hardware Acceleration of Data Processing with New Memory**
Top Conference	***Jie Zhang***, *Gyuyoung Park, David Donofrio, John Shalf, Myoungsoo Jung*
	26th IEEE International Symposium on High-Performance Computer Architecture
ISPASS	**Data Direct I/O Characterization for Future I/O System Exploration**
Major Conference	*Mohammad Alian, Yifan Yuan,* ***Jie Zhang****, Ren Wang, Myoungsoo Jung, Nam Sung Kim*

	The IEEE International Symposium on Performance Analysis of Systems and Software
CAL	**FastDrain: Removing Page Victimization Overheads in NVMe Storage Stack**
SCI Journal	***Jie Zhang**, Miryeong Kwon, Sanghyun Han, Nam Sung Kim, Mahmut Kandemir and Myoungsoo Jung*
	IEEE Computer Architecture Letters (CAL)

2019

HPCA	**FUSE: Fusing STT-MRAM into GPUs to Alleviate Off-Chip Memory Access Overheads**
Top Conference	***Jie Zhang**, Myoungsoo Jung, Mahmut Kandemir,*
	25th IEEE International Symposium on High-Performance Computer Architecture
IISWC	**Faster than Flash: An In-Depth Study of System Challenges for Emerging Ultra-Low**
Major Conference	**Latency SSDs**
	*Sungjoon Koh, Junkyeok Jang, Changrim Lee, Miryeong Kwon, **Jie Zhang**, Myoungsoo Jung,*
	The 2019 IEEE International Symposium on Workload Characterization
DAC	**FlashGPU: Placing New Flash Next to GPU Cores**
Top Conference	***Jie Zhang**, Miryeong Kwon, Hyojong Kim, Hyesoon Kim, Myoungsoo Jung,*
	The 56th Design Automation Conference (DAC), 2019
TPDS	**Exploring Fault-Tolerant Erasure Codes for Scalable All-Flash Array Clusters**
Top Journal	*Sungjoon Koh, **Jie Zhang**, Miryeong Kwon, Jungyeon Yoon, David Donofrio, Nam Sung Kim, Myoungsoo Jung,*
	IEEE Transactions on Parallel and Distributed Systems (TPDS)
NVMW	**Addressing Fast-Detrapping for Reliable 3D NAND Flash Design**
	*Mustafa Shihab, **Jie Zhang**, Myoungsoo Jung, Mahmut Kandemir,*
	10th Annual Non-Volatile Memories Workshop -- Nominated as Memorable Paper Award
KCC	**Maximizing GPU Cache Utilization with Adjustable Cache Line Management**
	***Jie Zhang**, Myoungsoo Jung,*
	Korean Computer Congress (KCC), 2019 -- Excellent Paper Award

2018

OSDI	**FlashShare: Punching Through Server Storage Stack from Kernel to Firmware for**
Top Conference	**Ultra-Low Latency SSDs**
	***Jie Zhang**, Miryeong Kwon, Donghyun Gouk, Changlim Lee, Mohammad Alian, Myoungjun Chun, Mahmut Kandemir, Nam Sung Kim, Jihong Kim, Myoungsoo Jung,*
	13th USENIX Symposium on Operating Systems Design and Implementation
MICRO	**Amber: Enabling Precise Full-System Simulation with Detailed Modeling of All SSD**
Top Conference	**Resources**

*Donghyun Gouk, Miryeong Kwon, **Jie Zhang**, Sungjoon Koh, Wonil Choi, Nam Sung Kim, Mahmut Kandemir, Myoungsoo Jung,*

The 51st Annual IEEE/ACM International Symposium on Microarchitecture

TACO **ReveNAND: A Fast-Drift Aware Resilient 3D NAND Flash Design**

Top Journal *Mustafa Shihab, **Jie Zhang**, Myoungsoo Jung, Mahmut Kandemir,*

ACM Transactions on Architecture and Code Optimization (TACO), 2018

Eurosys **FlashAbacus: A Self-governing Flash-based Accelerator for Low-power Systems**

Top Conference ***Jie Zhang**, Myoungsoo Jung,*

The European Conference on Computer Systems (EuroSys), 2018

IPDPS **CIAO: Cache Interference-Aware Throughput-Oriented Architecture and Scheduling for**

Major Conference **GPUs**

***Jie Zhang**, Shuwen Gao, Nam Sung Kim, Myoungsoo Jung,*

32nd IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2018

2017

CAL **SimpleSSD: Modeling Solid State Drive for Holistic System Simulation**

Major Journal *Myoungsoo Jung, **Jie Zhang**, Ahmed Abulila, Miryeong Kwon, Narges Shahidi, John Shalf, Nam Sung Kim and Mahmut Kandemir,*

IEEE Computer Architecture Letters (CAL), 2017

IISWC **Understanding System Characteristics of Online Erasure Coding on Scalable, Distributed**

Major Conference **and Large-Scale SSD Array Systems**

*Sungjoon Koh, **Jie Zhang**, Miryeong Kwon, Jungyeon Yoon, David Donofrio, Nam Sung Kim, Myoungsoo Jung,*

IEEE International Symposium on Workload Characterization (IISWC), 2017

IISWC **TraceTracker: Hardware/Software Co-Evaluation for Large-Scale I/O Workload**

Major Conference **Reconstruction**

*Miryeong Kwon, **Jie Zhang**, Gyuyoung Park, Wonil Choi, David Donofrio, John Shalf, Mahmut Kandemir, Myoungsoo Jung,*

IEEE International Symposium on Workload Characterization (IISWC), 2017

NPC **An In-depth Performance Analysis of Many-Integrated Core for Communication Efficient Heterogeneous Computing**

***Jie Zhang**, Myoungsoo Jung,*

IFIP International Conference on Network and Parallel Computing (NPC), 2017

IJPP **Enabling Realistic Logical Device Interface and Driver for NVM Express Enabled Full**

Major Journal **System Simulations**

*Donghyun Gouk, **Jie Zhang**, Myoungsoo Jung,* *IFIP International Conference on Network and Parallel Computing (NPC) and Invited for International Journal of Parallel Programming (IJPP), 2017*

2016

HPCA **DUANG: Fast and Lightweight Page Migration in Asymmetric Memory Systems**
Top Conference *Hao Wang, **Jie Zhang**, Gieseo Park, Sharmila Shridhar, Myoungsoo Jung, Nam Sung Kim,*
IEEE Symposium on High Performance Computer Architecture (HPCA), 2016

ASBD **A Study for Block-level I/O Trace Reconstruction on All-Flash Arrays**
*Miryeong Kwon, **Jie Zhang**, Gyuyoung Park, Myoungsoo Jung,*
Workshop on Architectures and Systems for Big Data (ASBD@ISCA), 2016

NVMSA **An In-Depth Study of Next Generation Interface for Emerging Non-Volatile Memories**
Major Conference *Wonil Choi, **Jie Zhang**, Shuwen Gao, Jaesoo Lee, Myoungsoo Jung, Mahmut Kandemir,*
IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA), 2016

INFLOW **ROSS: A Design of Read-Oriented STT-MRAM Storage for Energy-Efficient Non-Uniform Cache Architecture**
Jie Zhang, Miryeong Kwon, Chanyoung Park, Myoungsoo Jung, Songkuk Kim,
USENIX Workshop on Interactions of NVM/Flash with Operating Systems and Workloads

INFLOW **Couture: Tailoring STT-MRAM for Persistent Main Memory**
*Mustafa Shihab, **Jie Zhang**, Shuwen Gao, Josep Sloan, Myoungsoo Jung,*
USENIX Workshop on Interactions of NVM/Flash with Operating Systems and Workloads

2015

ASBD **CoDEN: A Hardware/Software CoDesign Emulation Platform for SSD-Accelerated Near Data Processing**
***Jie Zhang**, Damian Szmulewicz, Erick Macias, Myoungsoo Jung,*
The 5th Workshop on Architecture and System for Big Data (ASBD), 2015

PACT **NVMMU: Direct Solid State Disk Access for GPU-Accelerated Data Processing**
Top Conference ***Jie Zhang**, David Donofrio, John Shalf, Myoungsoo Jung,*
The 24th International Conference on Parallel Architecture and Compilation Techniques

ICCD **OpenNVM: An Open-Sourced FPGA-based NVM Controller for Low Level Memory**
Major Conference **Characterization**
***Jie Zhang**, Gieseo Park, David Donofrio, Mustafa Shihab, John Shalf and Myoungsoo Jung,*
The 33rd International Conference on Computer Design (ICCD), 2015

PACT-SRC **Integrating 3D Resisteive Memory Cache into GPGPU for Energy-Efficient Data Processing**
***Jie Zhang**, David Donofrio, John Shalf and Myoungsoo Jung,*
International Conference on parallel Architecture and Compilation Techniques (PACT) –

tel: +8619910031394|jiez@pku.edu.cn

ACM SRC 2nd Runner Award, 2015

FAST-WiP **Shared Non-Volatile Mmeory Cache for Energy-Efficient High Throughput GPU Computing**

Jie Zhang and Myoungsoo Jung,

USENIX Conference on File and Storage Technologies Working in Progress (FAST WiP), 2015

2014

HotStorage **Power, Energy, and Thermal Considerations in SSD-Based I/O Acceleration**

Jie Zhang, Myoungsoo Jung,

6th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage 14), 2014

PATENTS

- “Memory controlling device and computing device including the same”, Myoungsoo Jung, Donghyun Gouk, Miryeong Kwon, Sungjoon Koh, Jie Zhang, America (US20190171566A1)
- “Flash-based accelerator and computing device including the same”, Myoungsoo Jung, Jie Zhang, America (US10824341B2, US20180321859, US20170285968)
- “基于闪存的加速器和包含其的计算设备”, Myoungsoo Jung, Jie Zhang, China (CN107291424)
- “基于闪存的加速器及包括该加速器的计算设备”, Myoungsoo Jung, Jie Zhang, China (CN109460369)
- “Resistance switching memory-based accelerator”, Myoungsoo Jung, Gyuyoung PARK, Jie Zhang, America (US20180321880A1)
- “PARALLEL PROCESSING UNIT, COMPUTING DEVICE INCLUDING THE SAME, AND THREAD SCHEDULING METHOD”, Jie Zhang, Myoungsoo Jung, America (WO2018021620)
- “MEMORY CONTROL APPARATUS AND COMPUTING DEVICE INCLUDING SAME”, JUNG MYOUNGSOO, GOUK DONGHYUN, KWON MIRYEONG, KOH SUNGJOON, 정명수, JIE ZHANG, 국동현, 권미령, 고성준 장지에, Korea (KR1020180126267)
- “COMPUTING DEVICE, METHOD OF PROCESSING INPUT/OUTPUT REQUEST, AND RECORDING MEDIUM”, Jie Zhang, Myoungsoo Jung, Donghyun Gouk, Miryeong Kwon, Sungjoon Koh, America (pending)
- “FLASH-BASED COPROCESSOR”, Jie Zhang, Myoungsoo Jung, America (pending)
- “FLASH STORAGE DEVICE AND METHOD OF SCHEDULING PAGE VICTIMIZATION”, Jie Zhang, Myoungsoo Jung, America (pending)

EXPERIENCE

Research Assistant, Computer Architecture and Memory System Lab **Sep 2013 – May 2021**

- Cache and memory system optimization in GPGPU and multi-core system.
- Non-volatile memory (including Spin-transfer torque magnetic random-access memory and Phase Change Random Access Memory) characterization and optimization.
- Performance, power and thermal optimizations of Solid State Disk (SSD).

External Activities

Journal Paper Review/Sub-review

- IEEE Transactions on Computer
- ACM Transactions on Storage
- ACM Transactions on Architecture and Code Optimization

- ACM Transactions on Computer Systems
- IEEE Transactions on Parallel and Distributed Systems
- IEEE Computer Architecture Letters
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems

Conference Paper Review/Sub-review

- Eurosys’27
- OSDI’26
- MICRO’25’24’18 ’16
- ISCA’26’25’24’23
- HPCA’27’26’25’24’18 ’16
- ATC’25’23
- ISPASS’26’25
- USENIX ATC’23
- ChinaSys’24’22
- SAC’22
- HotStorage’20
- DAC’20 ’19
- NVMSA’17 ’16
- ICCD’19 ’18 ’17 ’15
- IPDPS’18 ’16
- DATE’19
- ASPLOS’23’22’19 ’18 ’17

Invited Talks and Presentations

- Invited talk, “ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis”, Intel Computational Storage Lab, 2020
- Presentation, “ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis”, ISCA, online, 2020
- Presentation, “DRAM-less: Hardware Acceleration of Data Processing with New Memory”, HPCA, San Diego, CA, 2020
- Presentation, “Scalable Parallel Flash Firmware for Many-core Architectures”, FAST, Santa Clara, CA, 2020
- Presentation, “FUSE: Fusing STT-MRAM into GPUs to Alleviate Off-Chip Memory Access Overheads”, HPCA, Washington DC, 2019
- Presentation, “FlashGPU: Placing New Flash Next to GPU Cores”, DAC, Las Vegas, NV, 2019
- Presentation, “Maximizing GPU Cache Utilization with Adjustable Cache Line Management”, Jeju, South Korea, 2019
- Presentation, “FlashShare: Punching Through Server Storage Stack from Kernel to Firmware for Ultra-Low Latency SSDs”, Carlsbad, CA, 2018
- Presentation, “FlashAbacus: A Self-governing Flash-based Accelerator for Low-power Systems”, Porto, Portugal, 2018
- Presentation, “CIAO: Cache Interference-Aware Throughput-Oriented Architecture and Scheduling for GPUs”, IPDPS, Vancouver, Canada, 2018
- Presentation, “An In-depth Performance Analysis of Many-Integrated Core for Communication Efficient Heterogeneous Computing”, NPC, Anhui, China, 2017

- Presentation, “ROSS: A Design of Read-Oriented STT-MRAM Storage for Energy-Efficient Non-Uniform Cache Architecture”, Inflow, Savannah, GA, 2016
- Presentation, “Couture: Tailoring STT-MRAM for Persistent Main Memory”, Inflow, Savannah, GA, 2016
- Presentation, “CoDEN: A Hardware/Software CoDesign Emulation Platform for SSD-Accelerated Near Data Processing”, ASBD, Portland, OR, 2015
- Presentation, “NVMMU: Direct Solid State Disk Access for GPU-Accelerated Data Processing”, PACT, San Francisco, CA, 2015
- Presentation, “Integrating 3D Resistive Memory Cache into GPGPU for Energy-Efficient Data Processing”, PACT SRC, San Francisco, CA, 2015
- Presentation, “OpenNVM: An Open-Sourced FPGA-based NVM Controller for Low Level Memory Characterization”, ICCD, New York city, NY, 2015
- Presentation, “Shared Non-Volatile Memory Cache for Energy-Efficient High Throughput GPU Computing”, FAST WiP, Santa Clara, CA, 2015
- Presentation, “Power, Energy, and Thermal Considerations in SSD-Based I/O Acceleration”, HotStorage, Philadelphia, PA, 2014

Teaching Experience

- Advanced Computer Architecture (Fall’26, Fall’25, Fall’24, Fall’23)
- Computer Architecture (Fall’26, Fall’25, Fall’24, Fall’23, Fall’22)
- Introduction of Computer System (Fall’26, Fall’25, Fall’24, Fall’23, Fall’22)
- IIT 3002 Operating Systems (Fall’15, Fall’16)
- IIT 6036 Computer Organization and Design (Fall’15, Fall’16)
- IIT 7024 Advanced System Architecture (Spring’16)

Honors

- 2025: HPCA Hall of Fame
- 2025: OlympusMons Award
- 2023: Intel Young Faculty Research Program
- 2022: ACM SIGCSE Rising Star Award
- 2021: Our storage-class memory research is selected as KAIST breakthrough 50 years
- 2020-2021: Korean BK21+ Scholarship
- 2020: HPCA travel grant
- 2019: Annual Non-Volatile Memories Workshop (NVMW) -- Nominated as Memorable Paper Award
- 2019: Korea Computer Congress (KCC) -- Best Presentation Paper Award
- 2018: OSDI travel grant
- 2015: ACM Student Research Competition 2nd Runner Award